EDGE AI-BASED VIBRATION ANOMALY DETECTION SYSTEM

A PROJECT REPORT

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of

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DECLARATION

I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously written by another person nor material which has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that the report entitled "**EDGE AI BASED VIBRATION ANOMALY DETECTION SYSTEM**", *submitted by* **JITHIN PR**, **JOANNE ALICE THOMAS**, **SANJANA LAKSHMI S**, **SMIJADAS K S** *to Muthoot Institute of Technology and Science, Varikoli for the award of the degree of Bachelor of Technology in Computer Science & Engineering is a bonafide record of the project work carried out by her, under our supervision and guidance. The content of the report, in full or parts, has not been submitted to any other Institute or University for the award of any other degree or diploma.*

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ABSTRACT

The construction phase of infrastructure projects is highly susceptible to safety risks due to temporary structural configurations, continuous modifications, and dynamic operational activities. Conventional safety monitoring techniques primarily rely on periodic manual inspections and predefined threshold-based systems, which are inadequate for capturing real-time structural behaviour. Consequently, gradual or sudden anomalies often go undetected until they develop into critical failures, posing serious risks to both workers and infrastructure.

Structural anomalies are frequently reflected through changes in vibration characteristics such as amplitude, frequency, and overall response patterns. Therefore, vibration-based monitoring serves as an effective and practical approach for identifying abnormal structural behaviour at an early stage.

This project proposes an Edge AI-based vibration anomaly detection system for construction-phase safety monitoring using accelerometer data. In this system, accelerometers mounted on a scaled experimental structure continuously acquire tri-axial vibration signals generated under normal and simulated construction conditions. These time-series signals represent the dynamic response of the structure to varying operational and environmental influences.

The collected data is processed locally on an edge device, where an unsupervised machine learning model is employed to learn normal vibration patterns and detect deviations in real time. By performing inference at the edge, the system ensures low latency, reduced dependency on cloud infrastructure, and continuous operation even in environments with limited connectivity.

Upon detection of abnormal vibration patterns, the system generates immediate alerts through a buzzer mechanism, enabling timely intervention and enhancing on-site safety. The proposed approach offers a cost-effective, scalable, and efficient solution for real-time structural health monitoring during the construction phase.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The construction phase of infrastructure projects is inherently prone to safety risks due to continuously evolving site conditions, temporary structural configurations, and ongoing dynamic activities. Unlike completed structures, partially built systems are subjected to varying loads, material inconsistencies, and environmental disturbances, making them highly vulnerable to instability and unexpected failures.

During construction, structures experience continuous changes in geometry, load distribution, and external influences such as wind, machinery operation, and human activity. These factors generate complex vibration patterns within the structure. If not monitored effectively, abnormal structural responses may develop gradually or occur suddenly, leading to hazardous situations and potential accidents.

Traditional construction safety practices primarily rely on periodic manual inspections and visual assessments conducted by engineers and site supervisors. Although these methods provide basic safety assurance, they are limited in their ability to deliver continuous, real-time insights into structural behaviour. Human observation is often subjective and may fail to detect subtle changes or hidden anomalies, especially in large-scale or fast-paced construction environments.

Structural anomalies are frequently reflected through variations in vibration characteristics, including changes in amplitude, frequency, and response patterns. Monitoring these vibration signals provides a reliable and practical approach for identifying abnormal behaviour at an early stage. Continuous vibration monitoring enables the detection of structural irregularities before they escalate into critical failures.

In this context, the integration of Edge Artificial Intelligence (Edge AI) with vibration sensing offers a promising solution for real-time safety monitoring. Edge AI enables data processing and decision-making directly on local devices, eliminating the need for cloud dependency

and reducing latency. This ensures immediate analysis and faster response times, which are crucial in safety-critical environments such as construction sites.

This project proposes an Edge AI-based vibration anomaly detection system for construction-phase safety monitoring. The system utilizes accelerometer sensors to capture real-time vibration data from the structure. An unsupervised machine learning model is employed to learn normal vibration behaviour and identify deviations that indicate potential anomalies.

By processing data locally on an edge device, the system ensures continuous monitoring, low latency, and reliable performance even in environments with limited connectivity. Upon detecting abnormal vibration patterns, the system generates immediate alerts, enabling timely intervention and significantly enhancing construction-phase safety.

1.2 SCOPE AND MOTIVATION

The scope of this project is to develop a real-time structural safety monitoring system for the construction phase of infrastructure projects using vibration analysis. The system utilizes accelerometer sensors to continuously capture vibration data and processes it locally on an edge device. An unsupervised machine learning model is employed to learn normal vibration behaviour and detect anomalies in real time. Upon identifying abnormal vibration patterns, the system generates immediate alerts to ensure timely intervention. The proposed solution is designed to be cost-effective, efficient, and scalable, making it suitable for construction environments where structures are incomplete and continuously evolving.

The motivation for this project arises from the challenges associated with ensuring safety in construction sites, which are inherently dynamic and hazardous. Traditional safety monitoring methods rely on periodic manual inspections and visual assessments, which are limited in providing continuous real-time insights and are often prone to human error. These limitations make it difficult to detect subtle structural anomalies before they escalate into critical failures. Since structural irregularities are often reflected as changes in vibration characteristics such as amplitude and frequency, vibration-based monitoring provides a practical and reliable approach for early detection.

Furthermore, the integration of Edge Artificial Intelligence enables on-device data processing, reducing latency and eliminating dependence on cloud infrastructure. This ensures faster response times and reliable operation even in environments with limited connectivity. Therefore, this project is motivated by the need to develop a smart, real-time, and automated monitoring system that enhances construction safety, enables early anomaly detection, and minimizes risks associated with structural failures.

CHAPTER 2

PROPOSED WORK

2.1 OBJECTIVES

• To design and develop a real-time vibration monitoring system for construction-phase safety using suitable sensors and embedded hardware, ensuring continuous observation of structural behaviour.

• To acquire and process vibration data using accelerometer sensors, capturing tri-axis signals that represent the dynamic response of the structure under different operational conditions.

• To implement an unsupervised machine learning model that can learn normal vibration patterns of the structure and identify deviations without requiring labeled datasets.

• To deploy the anomaly detection model on a local backend server, while the ESP32 handles real-time data acquisition and communication.

• To detect abnormal structural behaviour in real time by analyzing changes in vibration characteristics such as amplitude, frequency, and response patterns.

•To generate immediate alerts (e.g., buzzer notification) when anomalies are detected, enabling early warning and timely intervention to prevent potential structural failures.

2.2 PROBLEM STATEMENT

The construction phase of infrastructure projects is characterized by continuously evolving structural configurations, dynamic loading conditions, and varying environmental influences. Unlike completed structures, partially built systems lack full stability and are subjected to irregular stresses caused by construction activities, machinery operations, and material variations. These factors significantly increase the

risk of structural instability, making construction sites highly vulnerable to accidents and failures. Ensuring safety during this phase is particularly challenging, as structural behaviour changes frequently and unpredictably.

Existing safety monitoring practices in construction primarily depend on periodic manual inspections and visual assessments carried out by engineers and site supervisors. While these methods provide a basic level of safety assurance, they are inherently limited in their effectiveness. Manual inspections are time-consuming, discontinuous, and prone to human error, making it difficult to detect subtle or hidden anomalies in real time. In many cases, abnormal structural responses develop gradually and remain unnoticed until they reach a critical stage, resulting in severe consequences such as structural damage, financial loss, or threats to human life.

Furthermore, modern monitoring systems that utilize advanced technologies often rely on cloud-based data processing and analysis. Although these systems offer improved analytical capabilities, they introduce new challenges such as high latency, dependency on stable internet connectivity, and increased computational costs. In construction environments, where connectivity may be unreliable and immediate response is crucial, such systems fail to provide timely alerts and rapid decision-making capabilities.

Structural anomalies are often reflected as variations in vibration characteristics, including changes in amplitude, frequency, and overall response patterns. However, there is a lack of efficient systems that can continuously monitor these vibration signals and intelligently interpret them in real time. This creates a significant gap in ensuring proactive safety measures during the construction phase.

Therefore, there is a critical need for a reliable, real-time, and automated monitoring system that can continuously analyze structural behaviour and detect anomalies at an early stage. Such a system should be capable of operating independently of cloud infrastructure, providing low-latency responses, and generating immediate alerts to enable timely intervention. Addressing this need is essential for enhancing construction safety, reducing risks, and preventing potential structural failures.

2.3 EXISTING SYSTEM AND PROPOSED SOLUTION

2.3.1 EXISTING SYSTEM

The existing systems used for safety monitoring during the construction phase primarily rely on manual inspection methods and basic sensor-based threshold monitoring. In most construction sites, engineers and supervisors conduct periodic inspections to assess structural stability and identify visible defects. While these methods are simple and widely adopted, they are limited in their ability to provide continuous monitoring and real-time analysis. Manual inspections are time-consuming, subjective, and prone to human error, making it difficult to detect subtle or hidden structural anomalies that may develop gradually over time.

In recent years, some advanced monitoring systems have incorporated sensor technologies to measure parameters such as vibration, strain, and displacement. However, many of these systems depend on predefined threshold values to identify abnormal conditions. Such threshold-based approaches are not always reliable, as they fail to adapt to varying construction environments and dynamic structural behaviour. Additionally, several modern systems rely on cloud-based data processing, where sensor data is transmitted to remote servers for analysis. This introduces challenges such as high latency, dependence on stable internet connectivity, and increased operational costs.

Due to these limitations, existing systems are often ineffective in providing immediate alerts and real-time decision-making capabilities, which are critical for construction-phase safety. As a result, abnormal structural conditions may go unnoticed until they escalate into serious risks. These shortcomings highlight the need for a more intelligent, adaptive, and real-time monitoring solution that can operate efficiently in dynamic construction environments.

2.3.2 PROPOSED SYSTEM

The proposed system introduces an intelligent and real-time solution for construction-phase safety monitoring using vibration analysis and Edge Artificial Intelligence. Unlike traditional methods, this system continuously monitors structural behaviour by acquiring vibration data through accelerometer sensors mounted on the structure. The collected tri-axis vibration signals represent the dynamic response of the structure under varying operational and environmental conditions.

The system processes the acquired data locally on an edge device, such as an ESP32, eliminating the need for cloud-based computation. An unsupervised machine learning model is implemented to learn the normal vibration patterns of the structure during standard operating conditions. By establishing a baseline of normal behaviour, the system is capable of identifying deviations that indicate potential anomalies without requiring labeled training data. This makes the system adaptive and suitable for dynamic construction environments where conditions frequently change.

A key advantage of the proposed system is its ability to perform real-time anomaly detection with minimal latency. Since all computations are carried out on the edge device, the system ensures fast processing and immediate decision-making without relying on internet connectivity. When abnormal vibration patterns are detected, the system generates instant alerts through a buzzer mechanism, enabling early warning and timely intervention to prevent potential hazards.

Overall, the proposed system provides a cost-effective, efficient, and scalable solution for enhancing safety during the construction phase. By integrating vibration sensing with edge-based machine learning, it overcomes the limitations of existing systems and offers continuous, reliable, and intelligent structural monitoring.

CHAPTER 3

PROJECT DESIGN

3.1 SYSTEM ARCHITECTURE

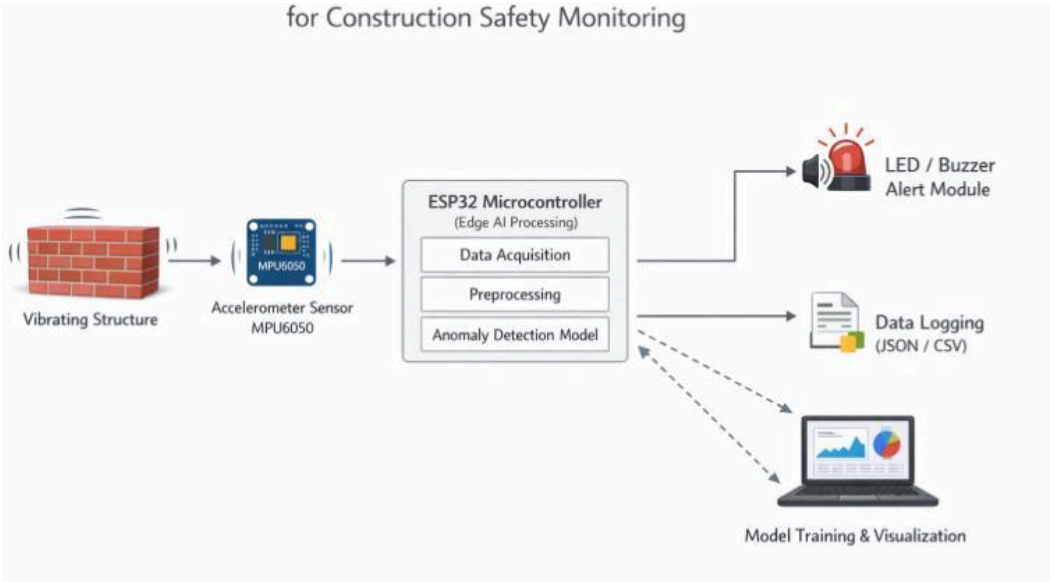


Figure 3.1: Architecture Diagram

The system architecture represents the overall workflow of the proposed vibration anomaly detection system. It consists of multiple components including the vibration sensor, edge processing unit, anomaly detection model, and alert mechanism.

Initially, the accelerometer sensor collects real-time vibration data from the structure in three axes. This data is then transmitted to the edge device (ESP32), where preprocessing and feature extraction are performed. The processed data is fed into an unsupervised machine learning model, which has been trained to recognize normal vibration patterns.

The model continuously analyzes incoming data and detects any deviation from normal behaviour. If an anomaly is identified, the system immediately triggers an alert through a buzzer or notification system. This ensures real-time monitoring and quick response, improving safety during the construction phase.

3.2 MODULES

3.2.1 HARDWARE MODULE

The hardware module forms the physical foundation of the proposed vibration anomaly detection system. It consists of sensors and embedded components that are responsible for collecting real-time vibration data and generating alerts based on structural conditions.

ESP32 DevKit V1 Microcontroller

The ESP32 DevKit V1 acts as the central control unit of the system. It is a high-performance dual-core 32-bit microcontroller operating at up to 240 MHz, with sufficient memory and processing capability to handle multiple tasks simultaneously. The ESP32 manages sensor data acquisition, preprocessing, machine learning inference, and alert generation in real time. Its built-in Wi-Fi and Bluetooth capabilities provide flexibility for future expansion such as remote monitoring or data transmission. The availability of multiple GPIO pins allows seamless interfacing with sensors and output devices.

MPU6050 Accelerometer Sensor

The MPU6050 is a widely used MEMS-based tri-axis accelerometer and gyroscope sensor. In this project, it is used primarily for vibration sensing. The sensor measures acceleration along the X, Y, and Z axes, capturing the dynamic behaviour of the structure. It communicates with the ESP32 via the I2C protocol, ensuring efficient and reliable data transfer. The high sensitivity of the sensor enables detection of even minor variations in vibration patterns.

Active Buzzer

An active buzzer is used as the alert mechanism in the system. It is directly controlled by the ESP32 through a digital output pin. When an anomaly is detected, the buzzer is activated to produce an audible warning signal. Since it is an active buzzer, it does not require complex signal generation, making it simple and effective for real-time alerts.

3.2.2 FIRMWARE MODULE

The firmware or processing module comprises all the software components programmed into the ESP32 microcontroller using the Arduino IDE. This module acts as the core intelligence of the system, coordinating data acquisition, signal preprocessing, anomaly detection, and alert generation in a continuous and synchronized manner. It ensures that the system operates efficiently in real time while maintaining reliability and accuracy in detecting abnormal structural behaviour.

Data Acquisition

The firmware or processing module comprises all the software components programmed into the ESP32 microcontroller using the Arduino IDE. This module acts as the core intelligence of the system, coordinating data acquisition, signal preprocessing, anomaly detection, and alert generation in a continuous and synchronized manner. It ensures that the system operates efficiently in real time while maintaining reliability and accuracy in detecting abnormal structural behaviour.

Data Preprocessing

The raw vibration data obtained from the sensor may contain noise and inconsistencies due to environmental disturbances and sensor limitations. To address this, preprocessing techniques such as noise filtering, smoothing, and normalization are applied. This step ensures that the data is clean, stable, and suitable for further analysis, thereby improving the overall accuracy and reliability of the anomaly detection process.

Anomaly Detection Logic

An unsupervised machine learning model is implemented within the firmware to learn the normal vibration behaviour of the structure. The model is trained using data collected under normal operating conditions and establishes a baseline pattern. During real-time operation, incoming data is continuously compared with the learned patterns. Any significant deviation beyond a predefined threshold is identified as an anomaly, indicating potential structural irregularities.

Real-Time Processing

All computations, including data acquisition, preprocessing, and anomaly detection, are performed locally on the ESP32. This ensures low-latency operation and eliminates dependency on external cloud infrastructure. The firmware is designed using non-blocking programming techniques, such as timer-based or `millis()`-based execution, allowing continuous data processing without interruption. This enables the system to respond instantly to abnormal conditions and maintain consistent monitoring performance.

3.2.3 EDGE AI MODULE

The Edge AI module is a crucial component of the proposed system, responsible for implementing machine learning algorithms on a local backend system (Flask server), which performs real-time analysis of sensor data received from the ESP32. Unlike conventional systems that rely on cloud-based processing, this module enables faster response times and improved system reliability. By integrating artificial intelligence at the backend, the system can continuously monitor structural behaviour and detect anomalies in real time without requiring external computational resources.

Model Selection

An unsupervised machine learning model, such as an autoencoder, is utilized for anomaly detection. This type of model is particularly suitable for the application, as it does not require labeled datasets, which are often difficult to obtain in construction environments. The model is trained using vibration data collected under normal operating conditions, allowing it to learn the inherent patterns and characteristics of standard structural behaviour.

Model Inference

Once trained, the model is deployed on a local backend server using Flask, while the ESP32 is responsible for data acquisition and communication. The vibration data collected by the ESP32 is continuously transmitted to the backend server, where it is fed into the trained model for inference. The model attempts to reconstruct the input data based on learned patterns, and the difference between the input and reconstructed output, known as the reconstruction error, is computed. If this error exceeds a predefined threshold, the

system classifies the input as abnormal, indicating potential structural issues.

Advantages

- Low latency processing
- No cloud dependency
- Real-time decision making
- Efficient resource utilization

3.2.4 ALERT AND NOTIFICATION MODULE

The Alert and Notification Module is responsible for providing immediate warnings when abnormal structural conditions are detected by the system. This module plays a critical role in ensuring safety by enabling quick response to potential hazards during the construction phase. It operates in real time and is directly triggered by the anomaly detection module whenever unusual vibration patterns are identified.

Buzzer Alert System

The system uses an active buzzer as the primary alert mechanism, which is connected to a GPIO pin of the ESP32 microcontroller. When the anomaly detection module identifies a deviation from normal vibration behaviour, the ESP32 generates a digital HIGH signal to activate the buzzer. Since an active buzzer is used, it produces sound instantly without requiring complex signal generation, making it highly suitable for real-time applications. The audible alert ensures that nearby workers and supervisors are immediately notified of potential structural issues.

Safety Response

The alert generated by the buzzer serves as an early warning system, allowing workers or site supervisors to take immediate corrective actions. This may include inspecting the structure, halting construction activities, or implementing preventive measures to avoid further risk. By providing instant feedback, the system helps in reducing the chances of structural failure, minimizing damage, and improving overall safety at the construction site.

Reliability and Real-Time Operation

Since the alert mechanism is integrated directly with the edge device, it does not depend on internet connectivity or external systems. This ensures reliable operation even in remote or low-connectivity environments. The low-latency response of the system guarantees that

alerts are generated without delay, making it highly effective for safety-critical applications.

3.2.5 DATA FLOW AND INTEGRATION MODULE

The Data Flow and Integration Module ensures smooth and efficient interaction between all components of the system. It defines how vibration data moves through different stages, from data acquisition to anomaly detection and alert generation. This module plays a crucial role in maintaining synchronization between hardware and software components, ensuring continuous and reliable system operation.

Working Flow

Initially, the accelerometer sensor captures real-time vibration data from the structure in three axes. This data is transmitted to the ESP32 microcontroller through the I2C communication protocol. The ESP32 then performs data preprocessing, including noise filtering and normalization, to prepare the data for analysis.

The processed data is fed into the machine learning model implemented on the edge device, where it is analyzed to detect any deviations from normal vibration behaviour. If the model identifies an anomaly based on predefined conditions or thresholds, the system immediately triggers the alert mechanism.

Finally, the alert module activates the buzzer to provide an audible warning, enabling quick response from workers or supervisors. This continuous flow of data ensures real-time monitoring and effective anomaly detection, improving the overall safety of the construction environment.

3.3 DATA FLOW DIAGRAM

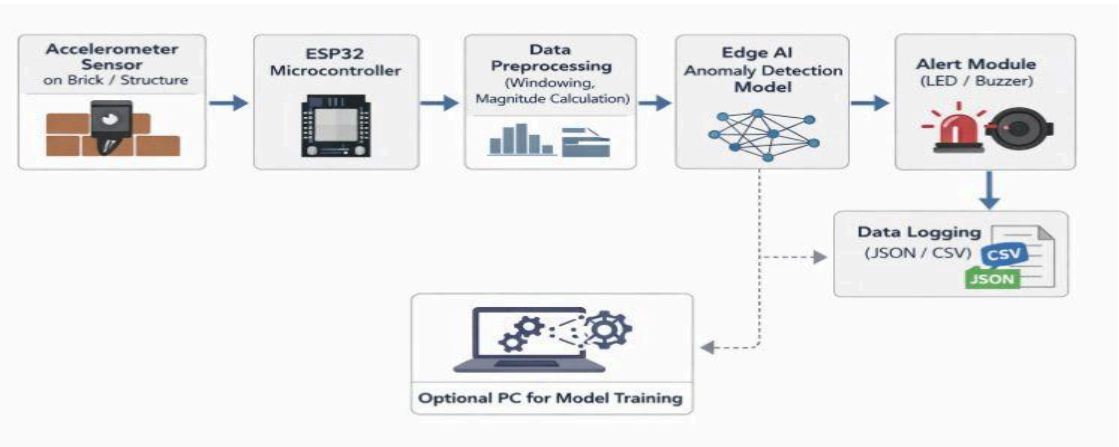


Figure 3.2: DF Diagram

The Data Flow Diagram (DFD) illustrates the flow of data within the system, starting from data acquisition to final output generation. The system begins with the MPU6050 sensor collecting real-time vibration data, which is then processed by the ESP32 microcontroller. The ESP32 sends sensor data to the backend server, where anomaly detection is performed, and results are returned. The processed data is transmitted to the backend server via HTTP communication.

The backend system performs further processing and applies the machine learning model to detect anomalies based on reconstruction error. The result is then sent back to the ESP32, which triggers the buzzer if an abnormal condition is detected. Additionally, the processed data and system status are displayed on the dashboard for monitoring by users such as workers and supervisors.

3.4 GUI DESIGN

3.4.1 OVERVIEW

The Graphical User Interface (GUI) of the proposed vibration anomaly detection system is designed to provide a simple, intuitive, and user-friendly platform for monitoring structural behaviour. The interface focuses on clarity, real-time data visualization, and ease of interpretation, enabling users to quickly identify abnormal conditions without requiring technical expertise.

3.4.2 Design Principles

• Simplicity: The interface is designed with minimal elements to ensure users can easily understand vibration data and alerts without confusion.

• Consistency: Uniform color schemes, fonts, and layouts are maintained across all screens for a professional and cohesive appearance.

• Accessibility: Clear text, readable graphs, and visible indicators are used to ensure information is easily accessible to all users.

• Responsiveness: The GUI is designed to adapt to different devices and screen sizes, ensuring smooth interaction.

3.4.3 KEY SCREENS

Worker Dashboard

This interface is designed for on-site workers. It displays real-time vibration data (X, Y, Z values) and system status. In case of abnormal conditions, immediate alerts are shown along with buzzer activation, enabling workers to take quick action.

Supervisor Dashboard

This interface provides a higher-level overview for supervisors. It includes real-time monitoring, anomaly alerts, and historical data analysis. Supervisors can observe trends, monitor multiple events, and make informed decisions regarding site safety.

Monitoring Screen

Displays live graphs of vibration signals and highlights abnormal patterns for better visualization.

Alert Screen

Shows warning messages and visual indicators when anomalies are detected.

Data Log / History Screen

Stores past vibration data and anomaly records for future analysis.

3.4.4 COLOUR SCHEME & VISUALIZATION

The GUI uses a simple and effective colour scheme to enhance readability and quick decision-making. Normal conditions are represented using calm colours such as green or blue, while abnormal conditions are highlighted using red or orange to draw immediate attention. Graphical representations such as line charts are used to visualize vibration data over time, making it easier to detect variations and anomalies.

3.5 TECHNOLOGY STACK

3.5.1 HARDWARE COMPONENTS

ESP32 Microcontroller

The ESP32 microcontroller serves as the central processing unit of the system. It is responsible for acquiring sensor data, performing real-time computations, and controlling the alert mechanisms. The ESP32 is chosen for its high processing capability, built-in WiFi support, and low power consumption, making it suitable for edge-based applications.

MPU6050 Sensor

The MPU6050 is a 3-axis accelerometer and gyroscope sensor used to capture vibration data along the X, Y, and Z axes. It communicates with the ESP32 using the I2C protocol, enabling efficient and continuous transmission of motion data for analysis.

Active Buzzer

An active buzzer is used to generate immediate audio alerts when abnormal vibration patterns are detected. It provides a simple and effective warning mechanism to notify workers and supervisors in real time.

3.5.2 SOFTWARE TECHNOLOGIES

Arduino IDE

Arduino IDE is used to program the ESP32 microcontroller. It enables real-time data acquisition from the vibration sensor and implements the processing logic required for anomaly detection and alert generation.

Python

Python is used for developing the machine learning model. It supports data preprocessing, training, and testing of the autoencoder model used for anomaly detection.

TensorFlow / Keras

TensorFlow and Keras are used to design and train the autoencoder neural network model for learning normal vibration patterns and identifying anomalies.

NumPy and Pandas

These libraries are used for efficient data handling, preprocessing, and numerical computations on vibration datasets.

Matplotlib

Matplotlib is used to visualize reconstruction error and threshold-based anomaly detection through graphs.

3.5.3 FRONT-END DEVELOPMENT

The front end of the system is designed to provide a simple and intuitive interface for monitoring vibration data and anomaly detection results. It enables both workers and supervisors to easily understand system status and respond to alerts in real time.

The user interface is developed to display key information such as real-time vibration values, anomaly status, and alert notifications. The system ensures clarity and ease of use, allowing users to quickly interpret data without technical complexity.

Key Features

Real-Time Data Display

Displays live vibration data (X, Y, Z axis values) received from the ESP32, allowing continuous monitoring of system conditions.

Anomaly Status Indicator

Shows whether the system is operating under normal or abnormal conditions based on the machine learning model output.

Alert Notifications

Provides visual and audio alerts when abnormal patterns are detected, ensuring immediate awareness.

Worker Dashboard

A simplified interface designed for on-site workers, showing current system status and alerts for quick action.

Supervisor Dashboard

Provides a more detailed view, including monitoring information and historical insights for better decision-making.

3.5.4 BACK-END DEVELOPMENT

The backend of the system is responsible for data acquisition, processing, anomaly detection, and communication between different system components. It ensures that vibration data collected from the sensor is analyzed efficiently and that appropriate actions are taken in real time.

The backend is implemented using a combination of the ESP32 microcontroller and a

Python-based Flask server. The ESP32 handles real-time data collection and alert control, while the Flask server performs machine learning inference and provides API communication.

Key Components

Data Acquisition Module

This module collects real-time vibration data from the MPU6050 sensor using the I2C communication protocol. The sensor provides acceleration values along the X, Y, and Z axes, which are continuously read by the ESP32 and transmitted for further processing.

Data Processing Module

The acquired sensor data is converted into a suitable format and organized into fixed-size windows for analysis. Basic preprocessing such as normalization and buffering is performed to ensure consistency between training data and real-time input.

Machine Learning Integration

An autoencoder-based anomaly detection model developed using Python and TensorFlow/Keras is integrated into the backend. The model compares incoming vibration data with learned normal patterns and computes reconstruction error to detect abnormal behavior.

Flask API Communication Module

A Flask-based backend server is used to handle communication between the ESP32 and the machine learning model. The ESP32 sends HTTP requests to the API, and the server processes the data, performs anomaly detection, and returns the result in JSON format.

Serial Communication Module

The backend reads sensor values from the ESP32 through serial communication using the PySerial library. The received data is stored in a buffer and used to form input windows for the anomaly detection model.

Alert Generation Module

Based on the anomaly detection result, the backend sends a response to the ESP32 indicating whether the system is in safe, warning, or danger state. If the error exceeds a predefined threshold, the ESP32 activates the buzzer to provide an immediate alert.

Analytics and Monitoring Module

The backend maintains a history of error values to calculate statistics such as average error, maximum error, and number of anomalies. These values are provided through API endpoints for monitoring and visualization.

3.6 SYSTEM REQUIREMENTS

3.6.1 OVERVIEW

The system requirements describe the hardware and software components needed to develop and run the proposed vibration anomaly detection system. The system is designed to run on an embedded platform with machine learning support and a monitoring dashboard.

3.6.2 HARDWARE REQUIREMENTS

Components	Specification	Purpose
ESP32 Microcontroller	Dual-core, WiFi enabled	Main processing unit
MPU6050 Sensor	3-axis accelerometer & gyroscope	Vibration data acquisition
Active Buzzer	5V buzzer	Alert generation
Connecting Wires	Standard jumper wires	Circuit connections
Breadboard / PCB	General purpose	Hardware assembly
Power Supply	5V USB / Adapter	System power
Computer / Laptop	Minimum 8GB RAM recommended	Model training & programming

3.6.3 SOFTWARE REQUIREMENTS

Software	Version	Purpose
Arduino IDE	2.x or above	ESP32 programming
Python	3.8+	Machine learning model
TensorFlow / Keras	Latest	Autoencoder model
NumPy	Latest	Numerical operations
Pandas	Latest	Dataset handling
Matplotlib	Latest	Graph plotting
VS Code / Python IDE	Any	Code development
GitHub	Online	Code storage
CSV File Support	—	Dataset storage

3.6.4 SYSTEM CONFIGURATION

Parameter	Requirement
Processor	Intel i3 / Ryzen 3 or higher
RAM	Minimum 8 GB
Storage	At least 5 GB free space
Operating System	Windows / Linux / macOS
Internet	Required for API communication & GitHub

3.6.5 FUNCTIONAL REQUIREMENTS

- The system shall collect vibration data from the sensor.
- The system shall process data using the ESP32 microcontroller.
- The system shall run the anomaly detection model.
- The system shall generate alerts when abnormal vibration occurs.
- The system shall display system status on the dashboard.
- The system shall support monitoring for both workers and supervisors.

3.6.6 NON FUNCTIONAL REQUIREMENTS

- Real-time data processing capability
- Low latency response
- Reliable and consistent alert generation
- User-friendly interface
- Low power consumption
- High accuracy in anomaly detection

3.6.7 STORAGE REQUIREMENTS

A minimum of 5 GB free storage space is required to store the machine learning model, dataset files, and necessary software libraries. This also supports the installation of development tools such as Arduino IDE and Python environments used in the system.

CHAPTER 4

IMPLEMENTATION

4.1 CODE SNIPPETS

Figure 4.1.1: *Initialization of ESP32, sensor, WiFi, and buzzer module*

```
void setup() {
  Serial.begin(115200);

  Wire.begin(21, 22);
  mpu.initialize();

  pinMode(buzzerPin, OUTPUT);
```

Figure 4.1.2: *Acquisition and conversion of vibration data from MPU6050 sensor*

```
int16_t ax, ay, az;
mpu.getAcceleration(&ax, &ay, &az);

float ax_g = ax / 16384.0;
float ay_g = ay / 16384.0;
float az_g = az / 16384.0;
```

Figure 4.1.3: *HTTP communication between ESP32 and backend server*

```
HTTPClient http;
http.begin(serverURL);

int code = http.GET();
```

Figure 4.1.4: *Decision logic based on server response for anomaly detection*

```
  if (payload.indexOf("true") > 0) {
    alertState = 1;
  } else {
    alertState = 0;
  }
}
```

Figure 4.1.5: *Buzzer control for alert generation based on anomaly status*

```
digitalWrite(buzzerPin, alertState ? HIGH : LOW);
```

Figure 4.1.6: *Creation of sliding windows for time-series vibration data*

```
window_size = 100
windows = []

for i in range(len(data) - window_size):
    windows.append(data[i:i+window_size])

windows = np.array(windows)
```

Figure 4.1.7: *Normalization of input data for consistent model training*

```
mean = windows.mean()
std = windows.std()

np.save("mean.npy", mean)
np.save("std.npy", std)

windows = (windows - mean) / std
```

Figure 4.1.8: *Autoencoder neural network architecture for anomaly detection*

```
model = Sequential([
    Dense(128, activation='relu', input_shape=(window_size * 3,)),
    Dense(64, activation='relu'),
    Dense(128, activation='relu'),
    Dense(window_size * 3)
])
```

Figure 4.1.9: *Training process of autoencoder model using vibration data*

```
model.fit(
    windows, windows,
    epochs=10,
    batch_size=32,
    validation_split=0.1
)
model.save("autoencoder_model.h5")
```

Figure 4.1.10: *Formation of input windows from continuous vibration data*

```
window_size = 100
windows = []

for i in range(len(data) - window_size):
    windows.append(data[i:i+window_size])

windows = np.array(windows)
```

Figure 4.1.11: *Normalization and reshaping of input data for model prediction*

```
windows_norm = (windows - mean) / std

windows_flat = windows_norm.reshape((windows_norm.shape[0], window_size * 3))
```

Figure 4.1.12: *Calculation of reconstruction error and threshold-based anomaly detection*

```
reconstructions = model.predict(windows_flat)

mse = np.mean((windows_flat - reconstructions) ** 2, axis=1)

threshold = 0.26
```

Figure 4.1.13: *Visualization of reconstruction error with anomaly threshold*

```
plt.plot(mse, label="Error")
plt.axhline(y=threshold, color='r', linestyle='--', label="Threshold")
```

Figure 4.1.14:*Loading trained autoencoder model and normalization parameters*

```
model = load_model("autoencoder_model.h5", compile=False)
mean = np.load("mean.npy")
std = np.load("std.npy")
```

Figure 4.1.15:*Processing of sensor data and computation of reconstruction error*

```
ow_norm = (window - mean) / std
ow_flat = window_norm.reshape(1, window_size

n = model.predict(window_flat, verbose=0)
= np.mean((window_flat - recon) ** 2)
```

Figure 4.1.16:*Classification of system status based on anomaly detection*

```
status = (
    "danger" if mse > THRESHOLD else
    "warning" if mse > THRESHOLD * 0.8 else
    "safe"
)
```

Figure 4.1.17*: API endpoint providing anomaly status to ESP32 for alert triggering*

```
@app.route("/api/alert")
def alert():
    if history and history[-1] > THRESHOLD:
        return "true"
    return "false"
```

Figure 4.1.18:*JSON response containing anomaly status and sensor information*

```
return jsonify({
    "error": round(float(mse), 3),
    "threshold": THRESHOLD,
    "status": status,
    "xyz": {
        "x": window[-1][0],
        "y": window[-1][1],
        "z": window[-1][2]
    },
    "location": location,
    "buzzer": mse > THRESHOLD
})
```

4.2 SCREENSHOTS

Figure 4.2.1:*Worker dashboard indicating safe system condition with no anomaly detected*

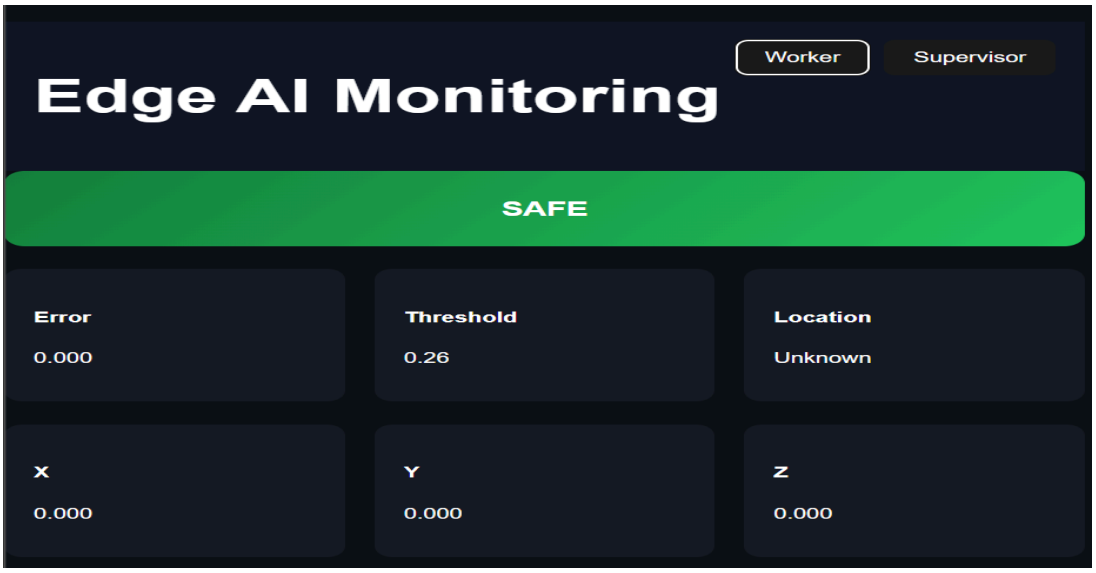


Figure 4.2.1:*Worker dashboard displaying anomaly detection with alert status during abnormal vibration*

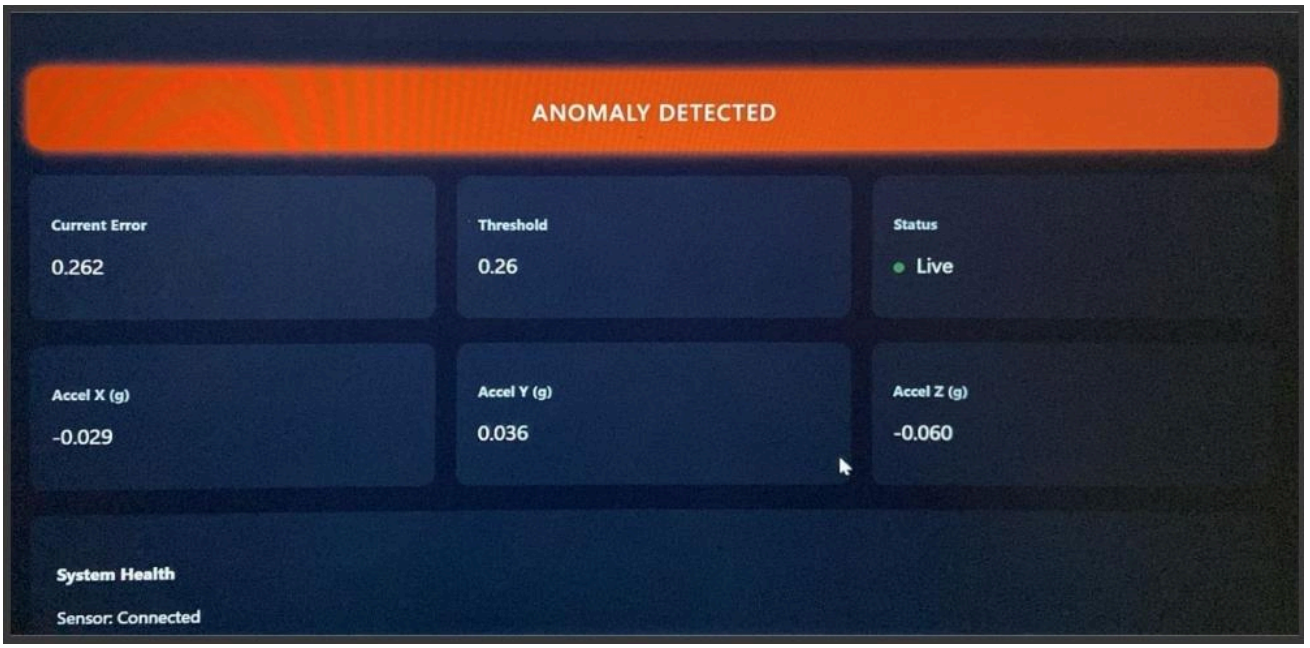
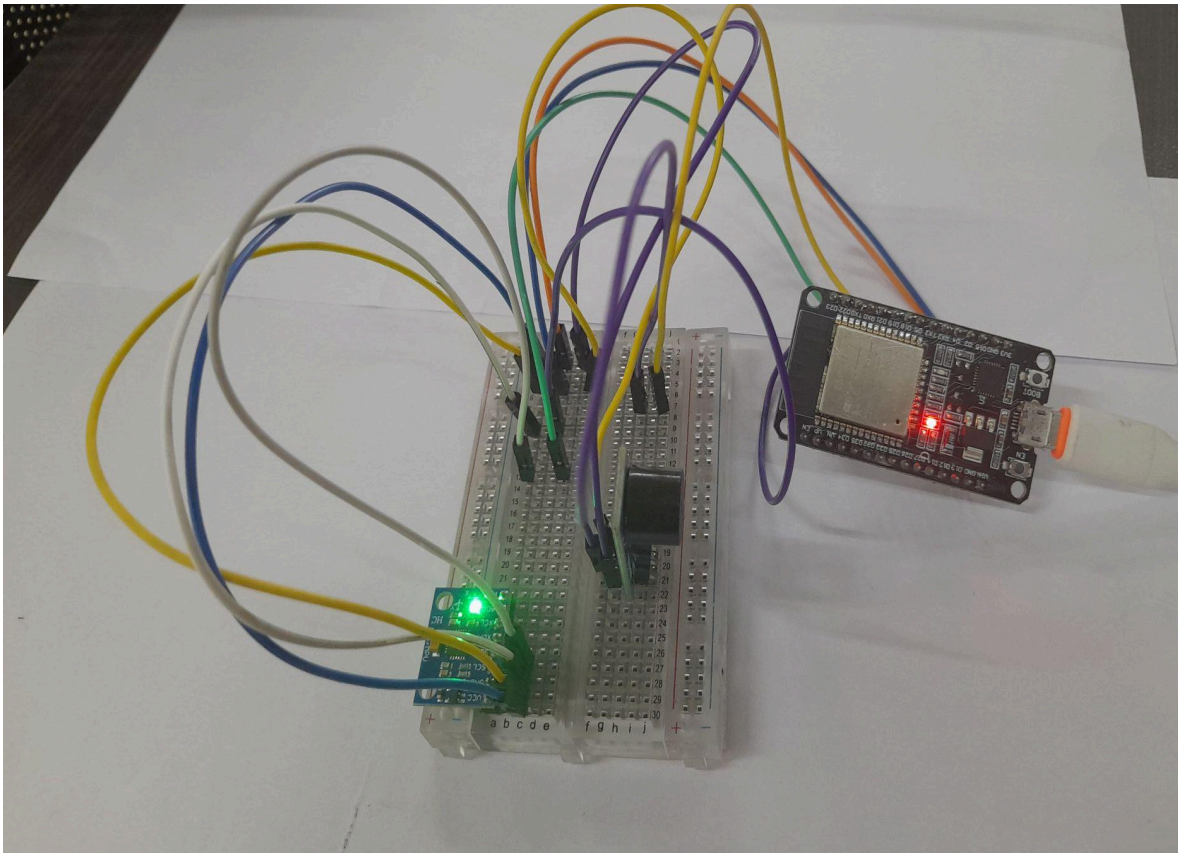


Figure 4.2.2:*Supervisor dashboard displaying reconstruction error graph with anomaly threshold and analytical metrics*

4.3 HARDWARE SETUP



CHAPTER 5
RESULTS AND ANALYSIS

Graph Output

The reconstruction error graph illustrates the variation of error values generated by the autoencoder model over time. A threshold line is defined to distinguish between normal and abnormal vibration patterns. When the reconstruction error remains below the threshold, the system operates under normal conditions. Any significant deviation above this threshold indicates potential anomalies in the structure.

Threshold Line

A predefined threshold value is used as a decision boundary for anomaly detection. This threshold is determined based on the model's performance during training. If the reconstruction error exceeds this threshold, the system classifies the input data as abnormal, ensuring accurate and reliable detection.

Sample Anomaly Detection

The system successfully detects anomalies when abnormal vibration patterns occur. During such instances, the reconstruction error exceeds the threshold value, and the system status changes accordingly. This is reflected in the dashboard, where anomaly conditions are clearly highlighted, enabling quick identification and response.

5.1 OBSERVATIONS

- The system effectively distinguishes between normal and abnormal vibration patterns with high reliability.
- The reconstruction error remains stable under normal operating conditions and shows a significant increase during anomalous events.
- Real-time communication between the ESP32 and the backend server ensures quick detection and response to abnormal conditions.
- The dashboard provides clear and intuitive visualization for both worker and supervisor roles, enhancing monitoring efficiency.
- The buzzer-based alert mechanism successfully notifies users immediately when

abnormal vibrations are detected.

5.2 ADVANTAGES OVER EXISTING SYSTEM

- Real-time monitoring:
  The system continuously monitors vibration data in real time, enabling immediate detection of abnormal conditions without delay.
- No manual inspection:
  Automated monitoring eliminates the need for manual inspection, reducing human effort and minimizing the chances of error.
- Adaptive ML model:
  The machine learning model adapts to normal vibration patterns, improving accuracy in detecting anomalies over time.
- Low latency alerts:
  The system provides instant alerts with minimal delay, ensuring quick response to potential structural issues.

5.3 SCOPE

- Cloud dashboard:
  The system can be extended to a cloud-based dashboard for remote monitoring and centralized data access.
  This enables real-time tracking of multiple construction sites from anywhere.
- Mobile app alerts:
  A mobile application can be developed to provide instant notifications to users.
  This ensures timely alerts and improved accessibility for workers and supervisors.
- Multiple sensors:
  The system can be scaled to support multiple sensors for large-area monitoring.
  This allows comprehensive structural analysis across different locations simultaneously.
- Predictive maintenance:
  Advanced models can be integrated to predict potential failures before they occur.
  This helps in proactive maintenance and reduces the risk of structural damage.

CHAPTER 6
CONCLUSION

The proposed Edge AI-based vibration anomaly detection system presents an effective and reliable solution for enhancing safety during the construction phase of infrastructure projects. By integrating accelerometer-based vibration sensing with edge computing, the system is capable of continuously monitoring structural behaviour and identifying abnormal patterns in real time. Unlike traditional safety approaches that rely on periodic manual inspections, the developed system provides continuous, automated monitoring, thereby reducing the risk of unnoticed structural issues.

The implementation of an unsupervised machine learning model enables the system to learn normal vibration behaviour without requiring labeled datasets, making it suitable for dynamic construction environments. The deployment of the model on the ESP32 edge device ensures low-latency processing, minimal dependence on external infrastructure, and efficient resource utilization. The inclusion of an immediate alert mechanism through a buzzer further strengthens the system by enabling quick response to potential hazards.

Overall, the system demonstrates a cost-effective, scalable, and practical approach to real-time structural health monitoring. It significantly improves safety by providing early warning of abnormal conditions and can be further extended with features such as remote monitoring, data logging, and advanced analytics. This project highlights the potential of Edge AI in transforming traditional safety practices into intelligent, automated systems for modern construction environments.

6.1 REFERENCES

1. Kim, S.-Y., & Mukhiddinov, M., *“Data Anomaly Detection for Structural Health Monitoring Based on a Convolutional Neural Network,”* Sensors, MDPI, 2023.
2. Malekloo, N., et al., *“Vibration-Based Structural Health Monitoring Using Machine Learning Techniques: A Review,”* Applied Sciences, MDPI, 2020.
3. Civera, M., & Surace, C., *“A Review of Vibration-Based Damage Detection Methods Using Machine Learning,”* Sensors, MDPI, 2019.